

TTK4240 Industriell elektroteknikk

February 10, 2025

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Lecture - Week –07 Transformers, Balanced Three-Phase Circuits

Transformers

- linear & Ideal

Balanced Three-Phase Voltages

- Three-phase voltage sources
- Generators
- Y-Y Circuits, Y- Δ Circuits

Power Calculations

- Average Power
- Complex Power
- Instantaneous Power

Measuring Average Power

- The Two-Wattmeter Method

Nilson's book Ch. 9 (9.10 and 9.11), Hambley book Ch. 15 (15.5 and 15.6)

Review of Magnetic Field Theory

Notions and Notations used in Magnetic Field Theory

Review

B Magnetic field or flux density

ψ Magnetic flux

$$\psi = \int \mathbf{B} \cdot d\mathbf{S}$$

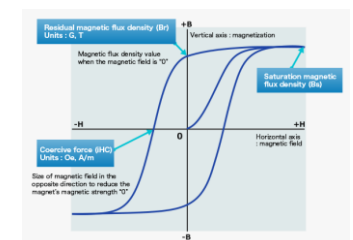
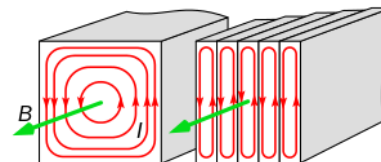
λ = $N\psi$ Fluxlinkage

H Magnetic field intensity

$$\mathbf{B} = \mu \mathbf{H} \quad (\text{ideal})$$

Practical magnetic materials have:

- Saturation;
- Hysteresis (loss);
- and Eddy current (loss).

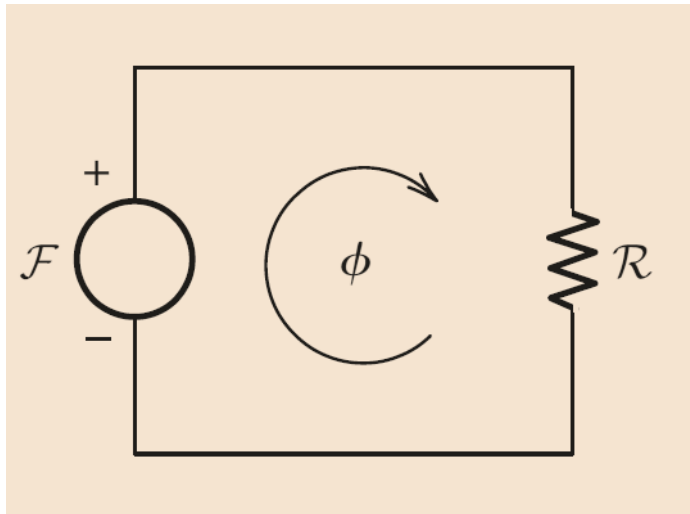


Physical laws in magnetic field theory

- **Lorentz's law**: defines the force imposed on a moving electric charge or a current carrying conductor. $\mathbf{F} = i\mathbf{l} \times \mathbf{B}$
- **Faraday's law**: indicates that the changing flux input to a coil can induce voltage in the coil, and **Lenz's law** further indicates the polarity of the induced voltage $e = -d\lambda / dt$
- **Ampere's law**: defines the magnetic field intensity surrounding current carrying conductors $\oint \mathbf{H} \cdot d\mathbf{l} = \Sigma i$

Magnetic circuit based on Ampere's law

Review



$$\mathcal{F} = Ni$$

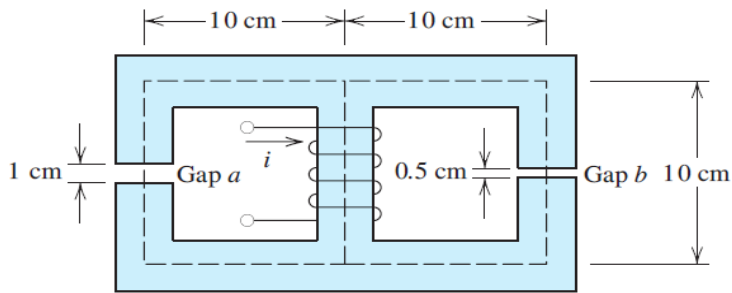
Magnetomotive force
(mmf)

$$\mathcal{R} = \frac{l}{\mu A}$$

Reluctance

$$\mathcal{F} = \mathcal{R}\phi$$

Ohm's law applies



Applies to any given magnetic systems,
the calculations and operations are
similar to electric circuits.

Magnetic coupling: self-inductance L and mutual-inductance M

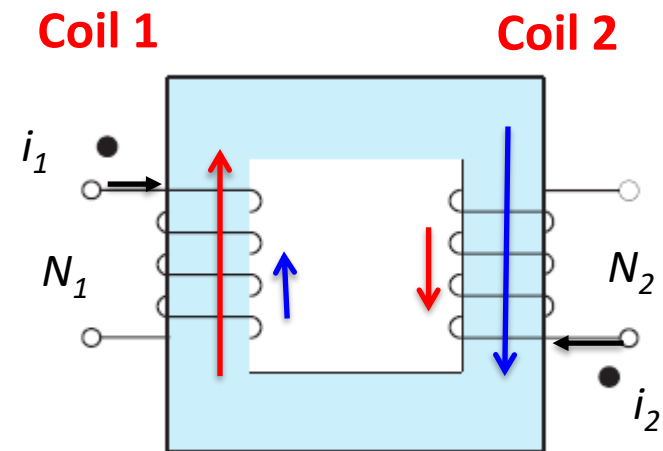
- Self-inductance is fluxlinkage per unit current of the same coil;
- Mutual inductance is the fluxlinkage of one coil created by per unit current from the coupled coil.

Self inductance

$$L_{11} = \frac{\lambda_{11}}{i_1} = \frac{N_1^2}{R_{core}}, L_{22} = \frac{\lambda_{22}}{i_2} = \frac{N_2^2}{R_{core}}$$

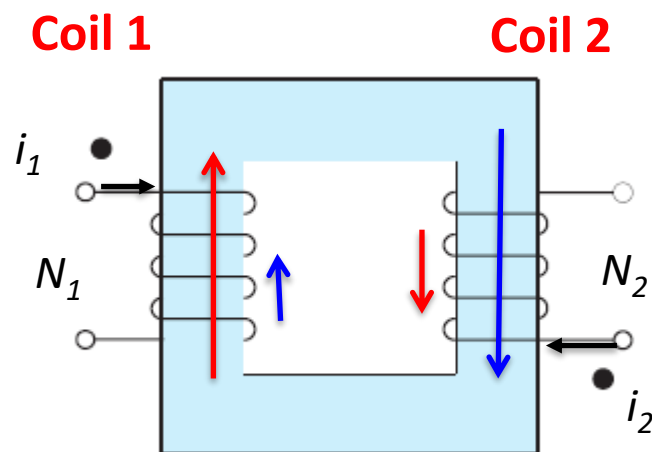
Mutual inductance

$$M_{12} = \frac{\lambda_{12}}{i_2} = M_{21} = \frac{\lambda_{21}}{i_1} = \frac{N_1 N_2}{R_{core}}$$



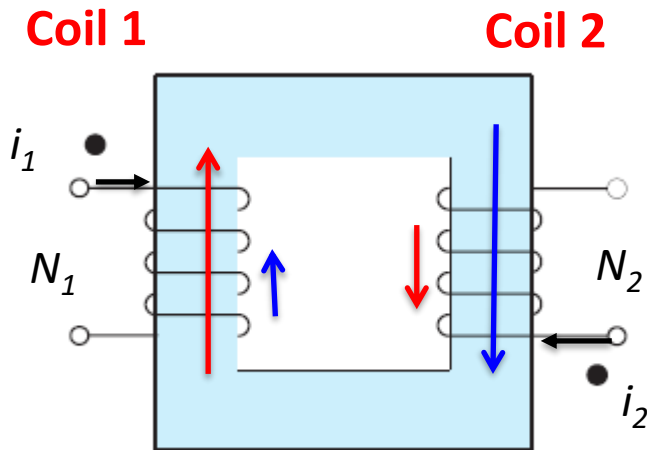
Dot convention

- The dot convention is used to simplify the determination of the **directions** of the **coupled fluxlinkage** and the **polarities** of the **mutually induced voltages**.



- if the currents of the coupled coils are flowing into the dots, the resulting fluxes will be in the same direction;
- If the current of one coil is flowing into its dot (**assuming is increasing**), the induced voltage on the other coil has a positive polarity at the corresponding dot.

Application of the Dot convention for Determination of Total Flux



What is the total flux for each of the coils?

$$\lambda_1 = \lambda_{11} + \lambda_{12} = L_{11}i_1 + M_{12}i_2$$

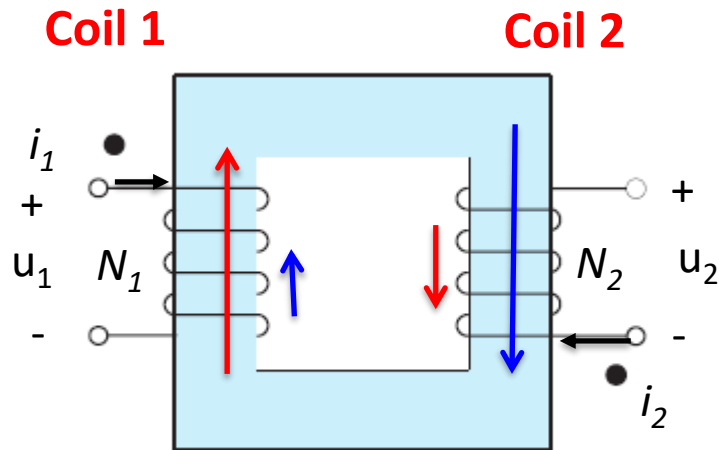
$$\lambda_2 = \lambda_{22} + \lambda_{21} = L_{22}i_2 + M_{21}i_1$$

$$M_{12} = \frac{\lambda_{12}}{i_2} = M_{21} = \frac{\lambda_{21}}{i_1} = \frac{N_1 N_2}{R_{core}}$$

$$L_{11} = \frac{\lambda_{11}}{i_1} = \frac{N_1^2}{R_{core}}, L_{22} = \frac{\lambda_{22}}{i_2} = \frac{N_2^2}{R_{core}}$$

Application of the Dot convention for Determination of Induced voltage

Review

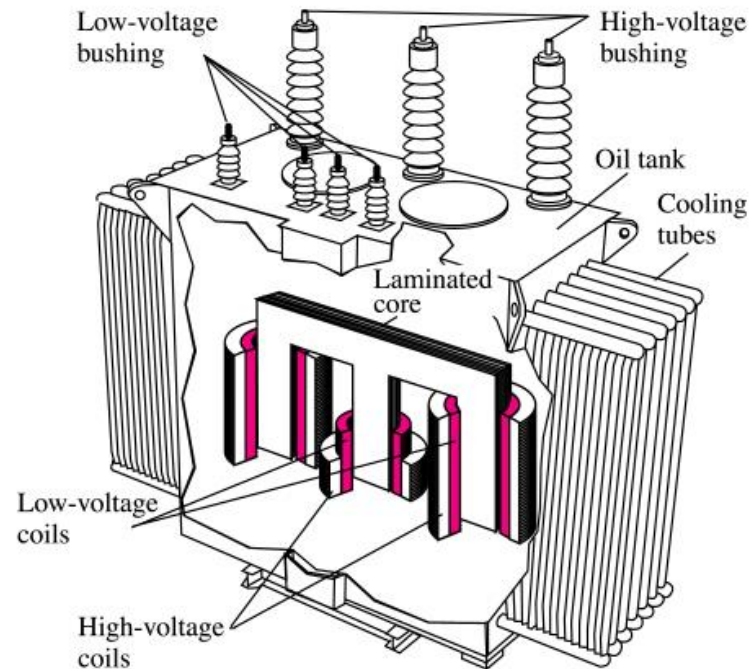


What is the total voltage for each of the coils?

$$u_1 = \underbrace{L_{11} \frac{di_1}{dt}}_{\text{Self-induced}} + \underbrace{M_{12} \frac{di_2}{dt}}_{\text{Mutual-induced, Positive because } i_2 \text{ flows into the dot, thus the induced voltage on the other coil is positive at the dot.}}$$

$$u_2 = \underbrace{-L_{22} \frac{di_2}{dt}}_{\text{Self-induced}} - \underbrace{M_{21} \frac{di_1}{dt}}_{\text{Mutual-induced, Positive because } i_1 \text{ flows into the dot}}$$

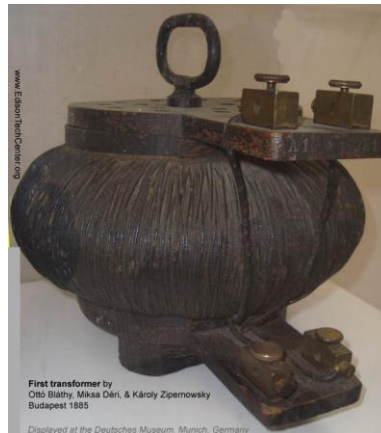
Transformers: Linear & Ideal



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What is a Transformer

Is a device that transfers electrical energy from one circuit to another through the magnetically coupled coils—the transformer's coils.

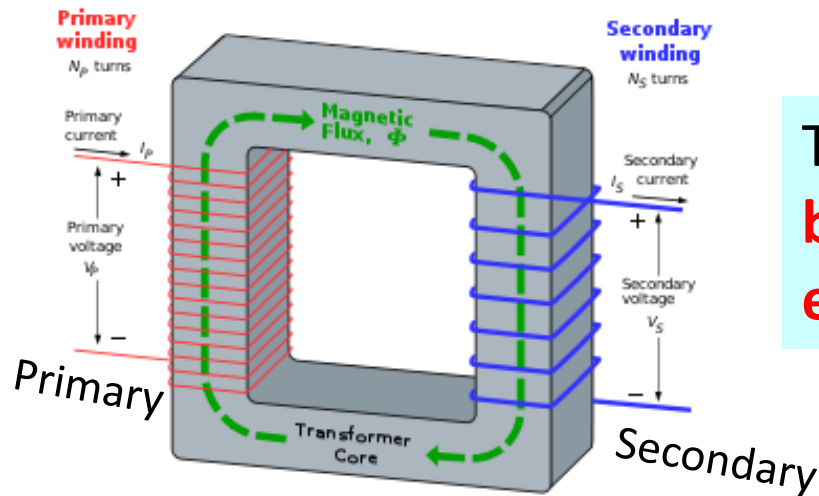


1878 -1883 - The Ganz Company (Budapest, Hungary) uses induction coils for their lighting systems with AC incandescent systems. This is the first appearance and use of the toroidal shaped transformer.



1885 - William Stanley makes the transformer more practical with some design changes: *it uses a soft iron core and adjustable gaps to regulate the amount of Electromagnetic Field in the secondary winding.*

How is it used?



The main function of transformer is **to bring voltage up or down in an AC electrical circuit**.

There are transformers all over every house, inside the black plastic case you plug into the wall to recharge your cell phone or other devices.

They can be **very large**, as in national utility systems, or it can be **very small** embedded inside electronics. **It is an essential part of all electronics today.**

<https://www.youtube.com/watch?v=U3CubKnkO4c>

Very
instructive
video!!

Working principle is based on :

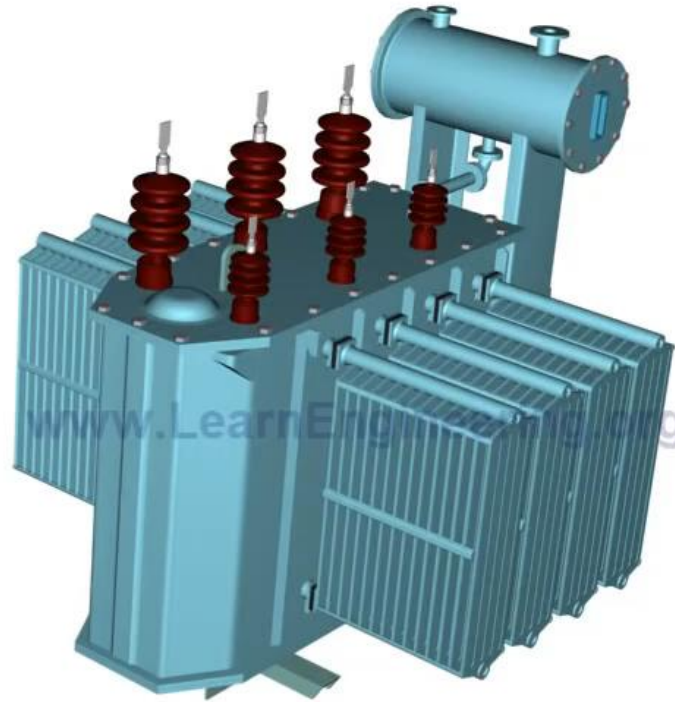
Electromagnetic Induction (Faraday-Lenz Law)

$$e = \frac{d\lambda}{dt} = N \frac{d\psi}{dt}$$

Electromagnetic Coupling

$$e = \frac{d\lambda_{12}}{dt} = M \frac{di_2}{dt} \quad \Leftrightarrow \quad e = \frac{d\lambda_{21}}{dt} = M \frac{di_1}{dt}$$

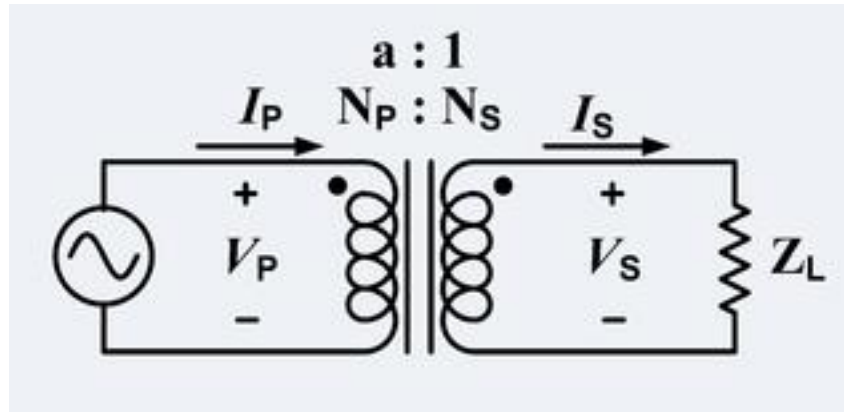
How does a Transformer work?



www.LearnEngineering.org

Ideal Transformer

Ideal Transformer

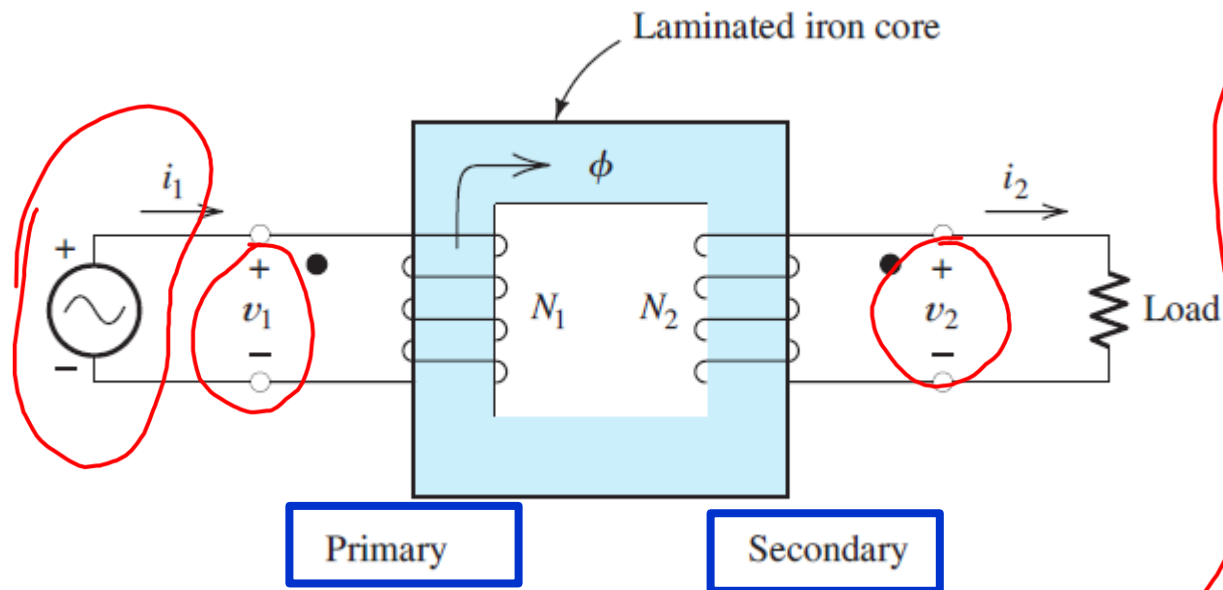


$$R = 0$$

An ideal transformer is a theoretical model meaning that

1. there are no energy losses and flux is completely confined within the magnetic core.
2. Perfect coupling implies infinitely high core magnetic permeability, which means the core reluctance is infinite small while the winding inductances is infinite large.

Voltage Ratio $\frac{V_2}{V_1} = \frac{N_2}{N_1}$



$$\phi(t) = \frac{V_{1m}}{N_1 \omega} \sin(\omega t)$$

$$v_2(t) = N_2 \frac{d\phi}{dt}$$

$$v_2(t) = N_2 \frac{V_{1m}}{N_1 \omega} \frac{d}{dt} [\sin(\omega t)]$$

$$v_2(t) = \frac{N_2}{N_1} V_{1m} \cos(\omega t)$$

$$v_1(t) = V_{1m} \cos(\omega t)$$

$$v_1(t) = V_{1m} \cos(\omega t) = N_1 \frac{d\phi}{dt}$$

$$V_{2rms} = \frac{N_2}{N_1} V_{1rms}$$

$$v_2(t) = \frac{N_2}{N_1} v_1(t)$$

Current Ratio

Zero net magnetomotiveforce

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$\mathcal{F} = N_1 i_1(t) - N_2 i_2(t)$$

$$\mathcal{F} = \mathcal{R} \phi$$

$$\mathcal{R} = 0$$

$$\mathcal{F} = N_1 i_1(t) - N_2 i_2(t) = 0$$

$$i_2(t) = \frac{N_1}{N_2} i_1(t)$$

$$I_{2\text{rms}} = \frac{N_1}{N_2} I_{1\text{rms}}$$

$$\frac{i_2}{i_1} = \frac{N_1}{N_2}$$

Power in an Ideal Transformer

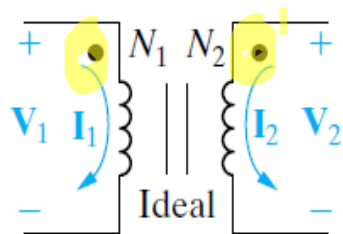
$$p_2(t) = v_2(t)i_2(t)$$

$$p_2(t) = \frac{N_2}{N_1}v_1(t)\frac{N_1}{N_2}i_1(t) = v_1(t)i_1(t)$$

$$p_2(t) = p_1(t)$$

In an ideal transformer power is not consumed, the primary side power equals the secondary side power

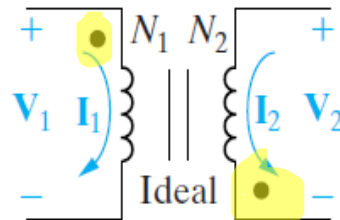
Dot convention for ideal transformer



$$\frac{\mathbf{V}_1}{N_1} = \frac{\mathbf{V}_2}{N_2},$$

$$N_1 \mathbf{I}_1 = -N_2 \mathbf{I}_2$$

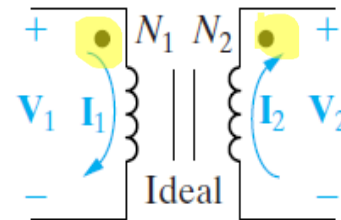
(a)



$$\frac{\mathbf{V}_1}{N_1} = -\frac{\mathbf{V}_2}{N_2},$$

$$N_1 \mathbf{I}_1 = N_2 \mathbf{I}_2$$

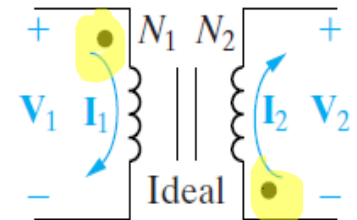
(b)



$$\frac{\mathbf{V}_1}{N_1} = \frac{\mathbf{V}_2}{N_2},$$

$$N_1 \mathbf{I}_1 = N_2 \mathbf{I}_2$$

(c)



$$\frac{\mathbf{V}_1}{N_1} = -\frac{\mathbf{V}_2}{N_2},$$

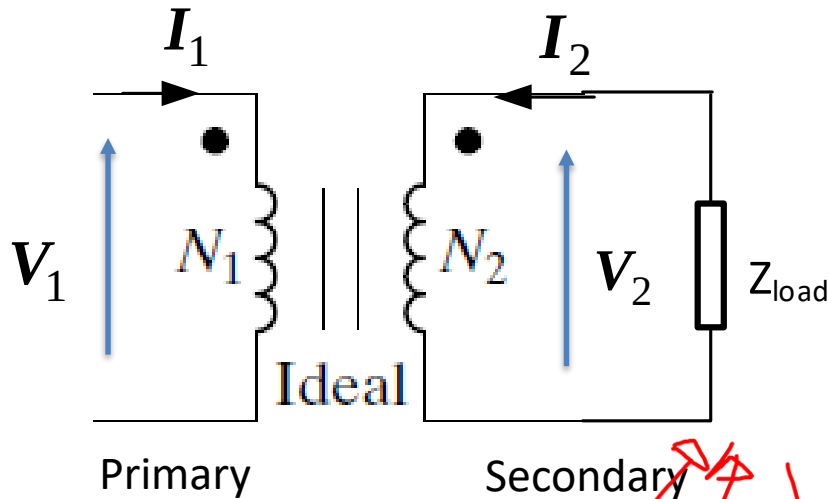
$$N_1 \mathbf{I}_1 = -N_2 \mathbf{I}_2$$

(d)

If the coil voltages $\mathbf{V}_1$ and $\mathbf{V}_2$ are both positive or negative at the dot-marked terminal, use a plus sign in Eq. 9.76. Otherwise, use a negative sign.

If the coil currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are both directed into or out of the dot-marked terminal, use a minus sign in Eq. 9.77. Otherwise, use a plus sign.

Impedance seen from the primary side



$$N_1 I_1 = -N_2 I_2$$

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

$$-\frac{N_1}{N_2} I_1 = I_2$$

$$\frac{N_2}{N_1} V_1 = V_2$$

$$-Z_{load} I_2 = V_2$$

Red handwritten annotations include circles around I_2 , V_2 , and Z_{load} , and arrows pointing from these terms to the final equation on the right.

primary side

$$\frac{V_1}{I_1} = Z_{load} \left(\frac{N_1}{N_2} \right)^2$$

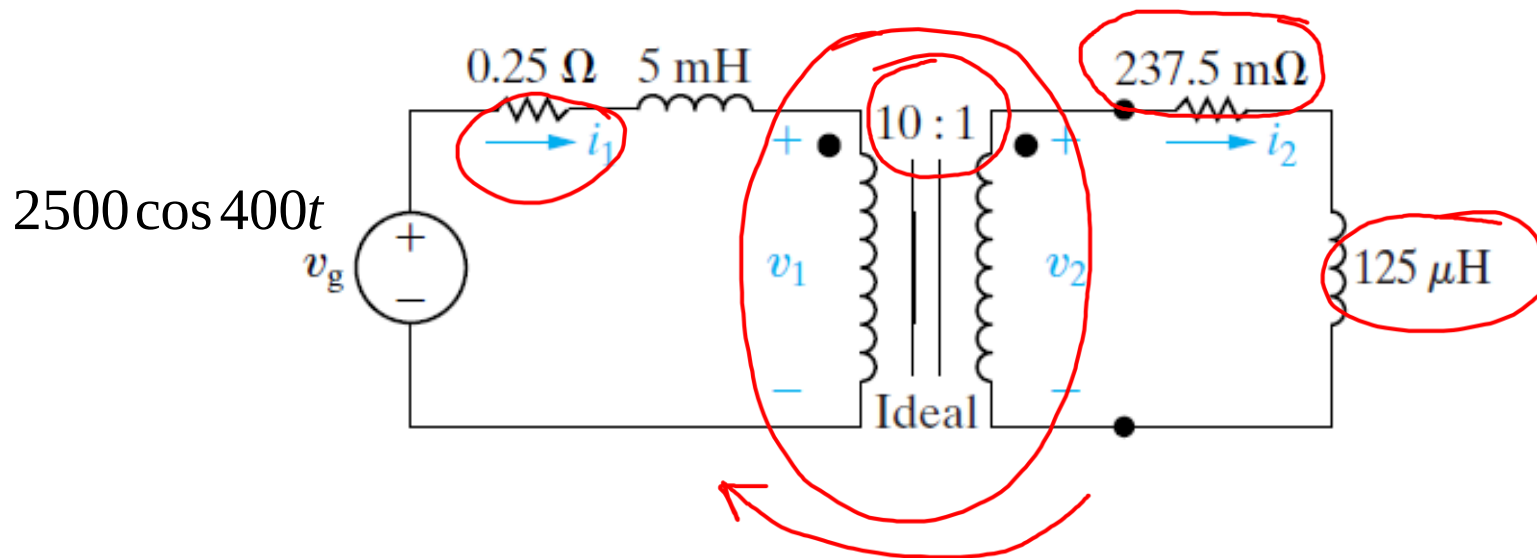
Impedance seen from primary side

Practical Example

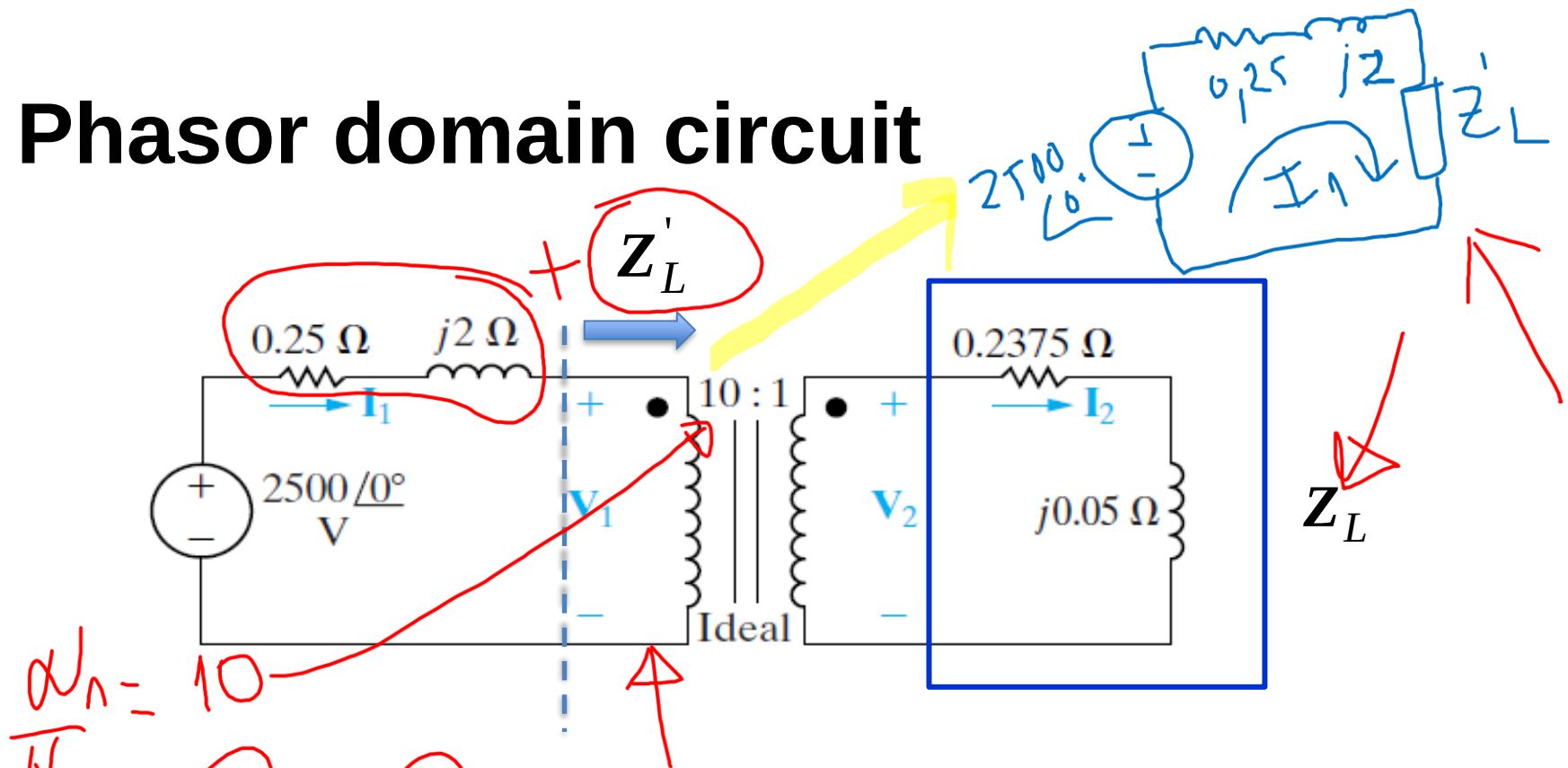
Ideal transformer: sinusoidal steady state analysis

The load impedance connected to the secondary winding of the ideal transformer in Fig. 9.45 consists of a $237.5 \text{ m}\Omega$ resistor in series with a $125 \text{ }\mu\text{H}$ inductor.

If the sinusoidal voltage source (v_g) is generating the voltage $2500 \cos 400t$ V, find the steady-state expressions for: (a) i_1 ; (b) v_1 ; (c) i_2 ; and (d) v_2 .



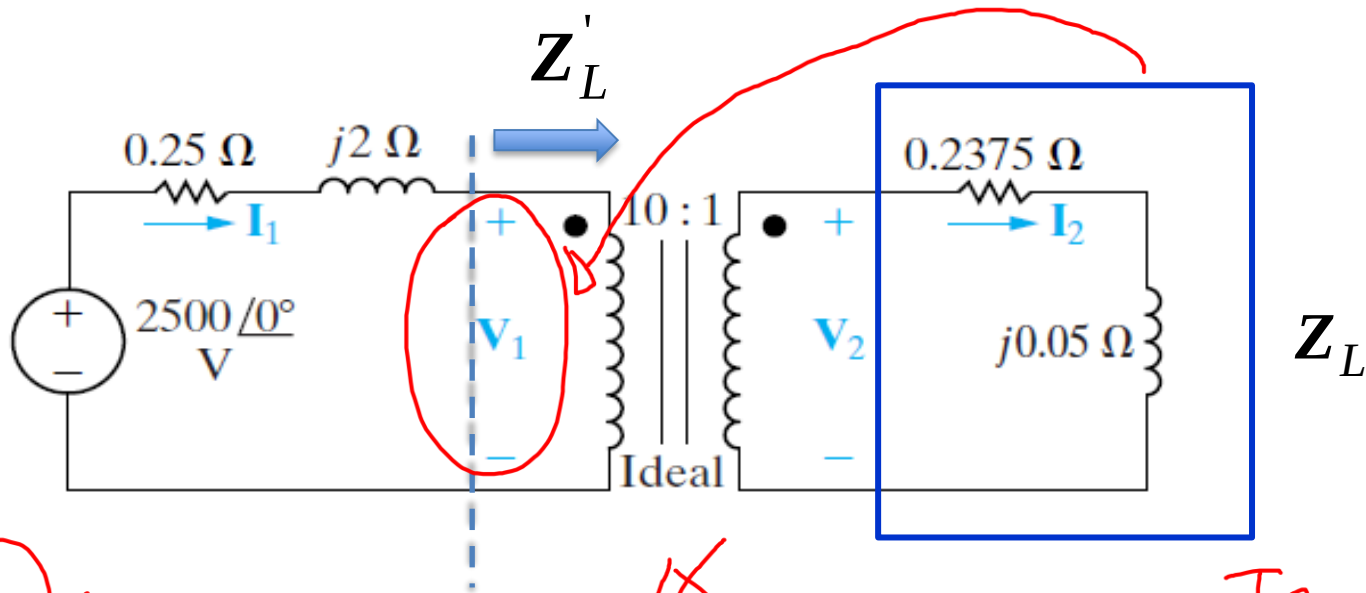
Phasor domain circuit



$$\frac{N_1}{N_2} = 10$$

$$\mathbf{Z}'_L = 10^2 \mathbf{Z}_L = 10^2 (0.2375 + 0.05j) = 23.75 + 5j = \underline{\underline{24.27 \angle 11.89^\circ}}$$

$$\mathbf{I}_1 = \frac{2500 \angle 0^\circ}{0.25 + 2j + \mathbf{Z}'_L} = \frac{2500 \angle 0^\circ}{0.25 + 2j + 23.75 + 5j} = \frac{2500 \angle 0^\circ}{25 \angle 16.26^\circ} = \underline{\underline{100 \angle -16.26^\circ}}$$



$$I_1 = 100 \angle -16.26^\circ$$

$$I_2 = 10 I_1 = 1000 \angle -16.26^\circ$$

$$\frac{I_2}{I_1} = \frac{N_1}{N_2} = 10$$

$$V_1 = Z'_L I_1 = 24.27 \angle 11.89^\circ \times 100 \angle -16.26^\circ = 2427 \angle -4.37^\circ$$

$$V_2 = V_1 / 10 = 242.7 \angle -4.37^\circ$$

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} = 1/10$$

$$i_1 = 100 \cos(400t - 16.26^\circ) \text{ A.}$$

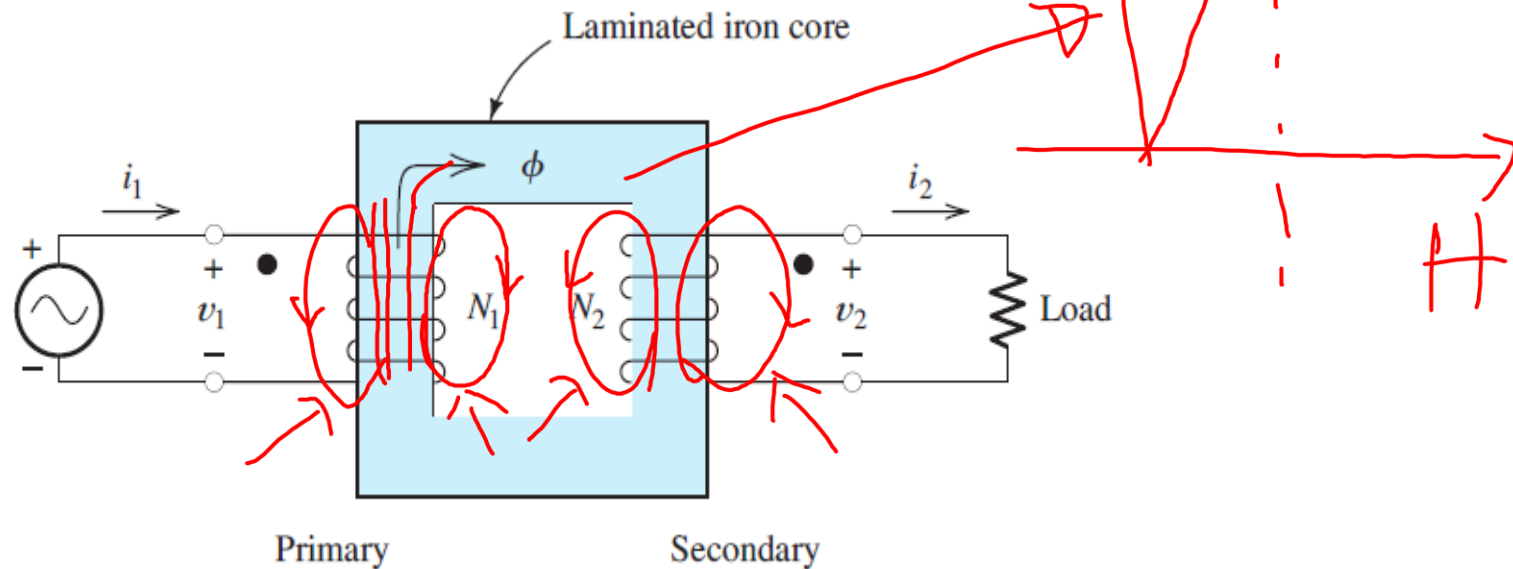
$$v_1 = 2427.06 \cos(400t - 4.37^\circ) \text{ V.}$$

$$i_2 = 1000 \cos(400t - 16.26^\circ) \text{ A.}$$

$$v_2 = 242.71 \cos(400t - 4.37^\circ) \text{ V.}$$

Linear Transformer

Linear transformer



- The main assumption is that the magnetic system is linear, given the name “linear transformer”
 - The core reluctance is not necessarily zero as assumed for the ideal transformer;
 - Losses due to resistance of the transformer can exist;
 - The flux is not assumed perfectly confined in the core, which means the Leakage inductance of the air can exist.

Differences from ideal transformer

- The **voltage ratio** of the ideal transformer holds for the linear transformer

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

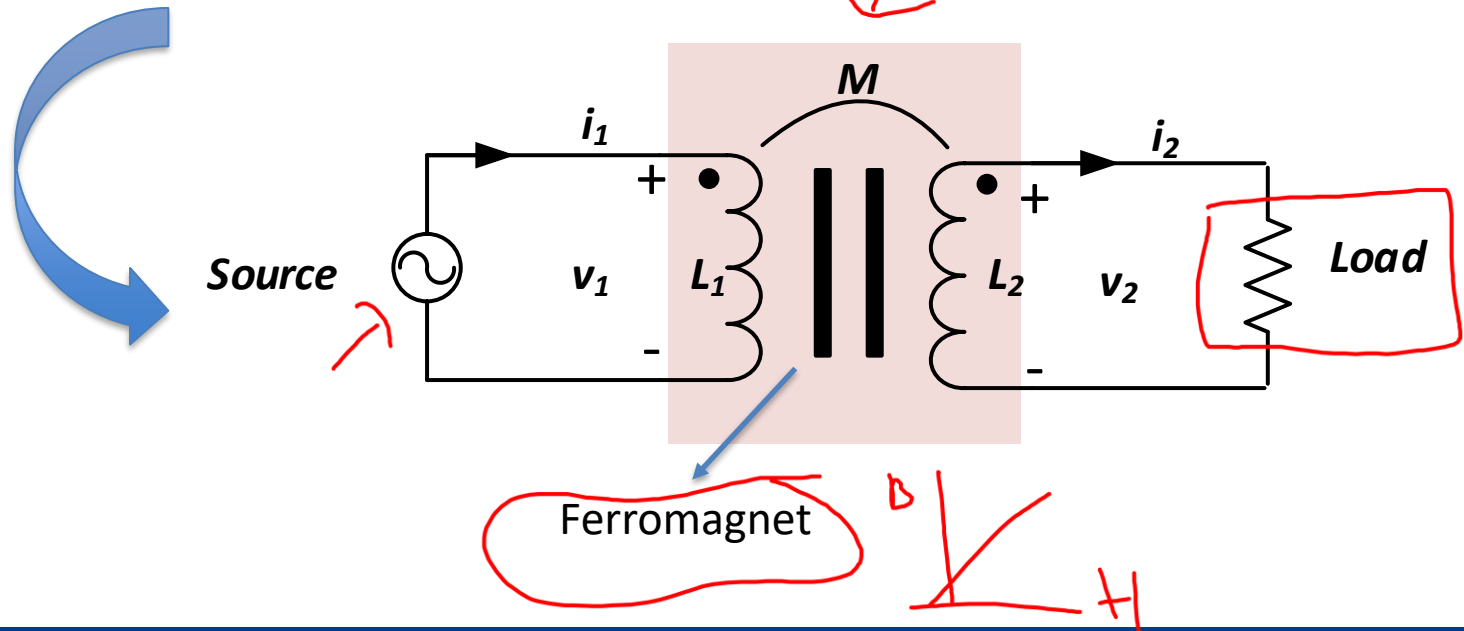
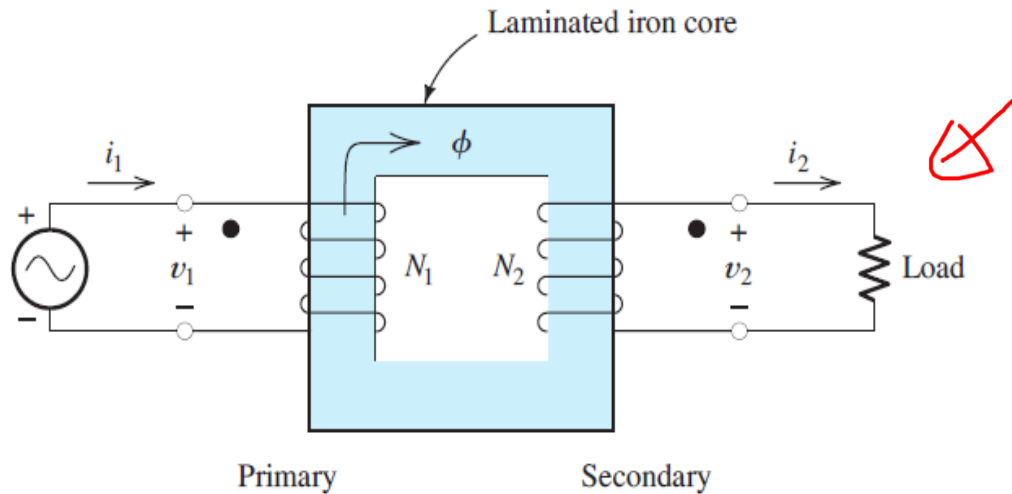
- The **current ratio and power balance** of the ideal transformer **DO NOT** hold for the linear transformer

$\frac{I_1}{I_2} \neq \frac{N_2}{N_1}$ $p_1 \neq p_2$ Instantaneous power is not balanced since the magnetic conductor stores energy

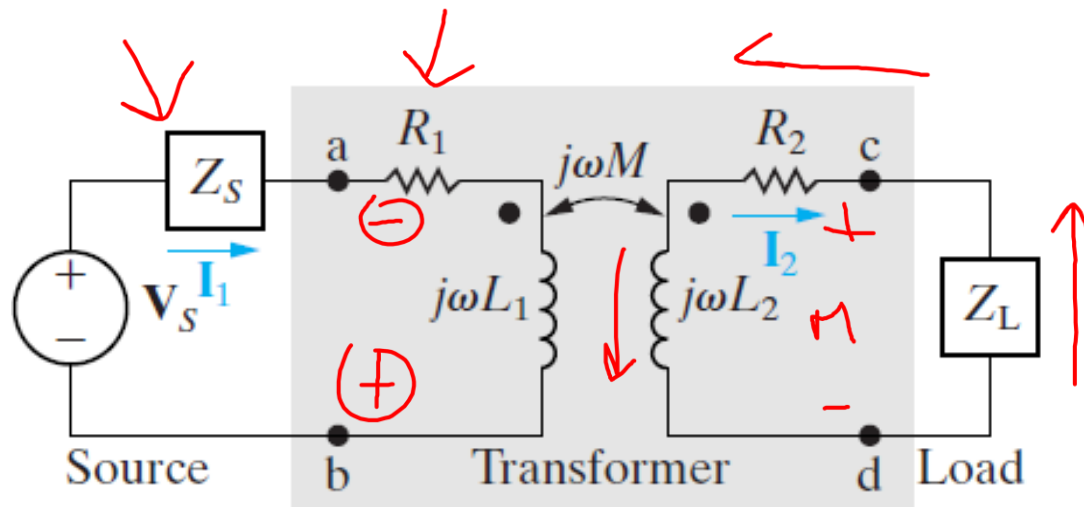
- The Impedance seen from the primary side **Does Not** hold as well

$\frac{V_1}{I_1} \neq Z_{load} \left(\frac{N_1}{N_2} \right)^2$

Linear Transformer: Time domain model



Linear Transformer: frequency/phasor domain model



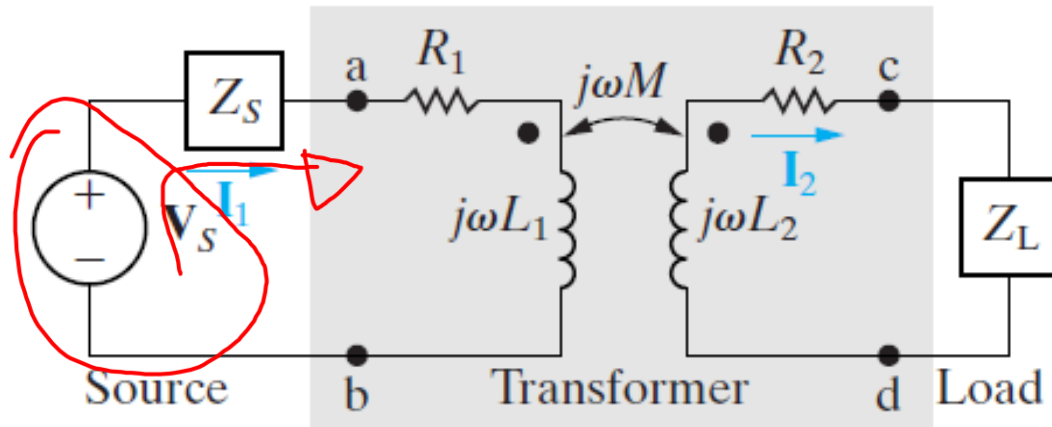
The analysis consists of finding:

- the current phasors I_1 and I_2 as a function of the circuit parameters.
- the impedance from the terminals ab

R_1 = the resistance of the primary winding,
 R_2 = the resistance of the secondary winding,
 L_1 = the self-inductance of the primary winding,
 L_2 = the self-inductance of the secondary winding,
 M = the mutual inductance.

$$\mathbf{V}_s = (Z_s + R_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2,$$

$$0 = -j\omega M\mathbf{I}_1 + (R_2 + j\omega L_2 + Z_L)\mathbf{I}_2.$$



Self impedances of primary and secondary sides:

$$Z_{11} = Z_s + R_1 + j\omega L_1,$$

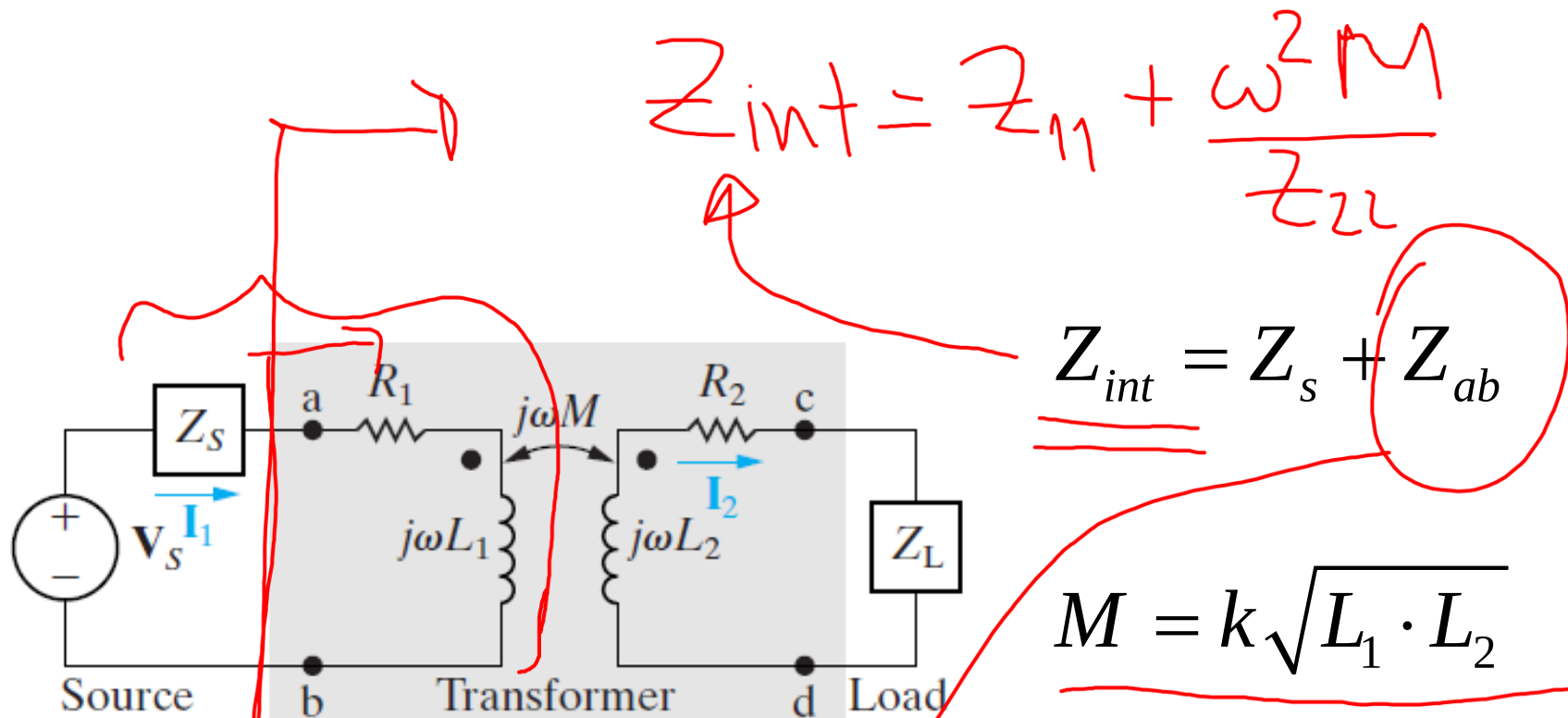
$$Z_{22} = R_2 + j\omega L_2 + Z_L,$$

$$\mathbf{I}_1 = \frac{Z_{22}}{Z_{11}Z_{22} + \omega^2 M^2} \mathbf{V}_s,$$

$$\mathbf{I}_2 = \frac{j\omega M}{Z_{11}Z_{22} + \omega^2 M^2} \mathbf{V}_s = \frac{j\omega M}{Z_{22}} \mathbf{I}_1.$$

To the internal source voltage, the impedance:

$$\frac{\mathbf{V}_s}{\mathbf{I}_1} = Z_{\text{int}} = \frac{Z_{11}Z_{22} + \omega^2 M^2}{Z_{22}} = Z_{11} + \frac{\omega^2 M^2}{Z_{22}}.$$



k: Coupling factor

$$Z_{ab} = Z_{11} + \frac{\omega^2 M^2}{Z_{22}} - Z_s = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{(R_2 + j\omega L_2 + Z_L)}$$

Reflected Impedance

Reflected impedance Z_r

$$\underline{Z_{ab}} = Z_{11} + \frac{\omega^2 M^2}{Z_{22}} - Z_s = \underbrace{R_1 + j\omega L_1} + \frac{\omega^2 M^2}{(R_2 + j\omega L_2 + Z_L)}.$$

$$\underline{Z_L = R_L + jX_L},$$

$$\underline{Z_{22} = R_2 + R_L + j(\omega L_2 + X_L)}.$$

Scaling factor

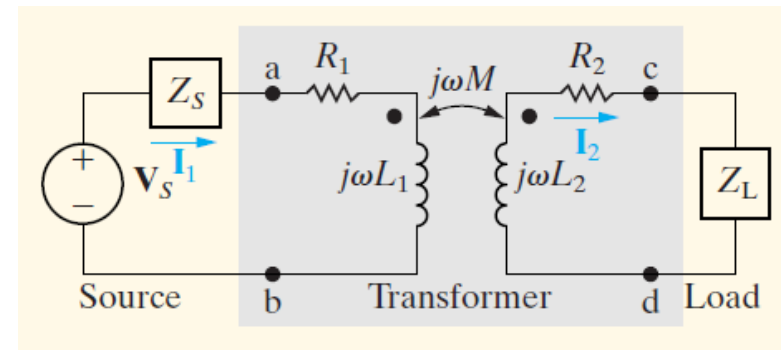
$$\begin{aligned} Z_r &= \frac{\omega^2 M^2}{R_2 + R_L + j(\omega L_2 + X_L)} \\ &= \frac{\omega^2 M^2 [(R_2 + R_L) - j(\omega L_2 + X_L)]}{(R_2 + R_L)^2 + (\omega L_2 + X_L)^2} \\ &= \frac{\omega^2 M^2}{|Z_{22}|^2} [(R_2 + R_L) - j(\omega L_2 + X_L)] \end{aligned}$$

$$Z_{22}^*$$

Practical Example

Example 9.13 Nilsson's book

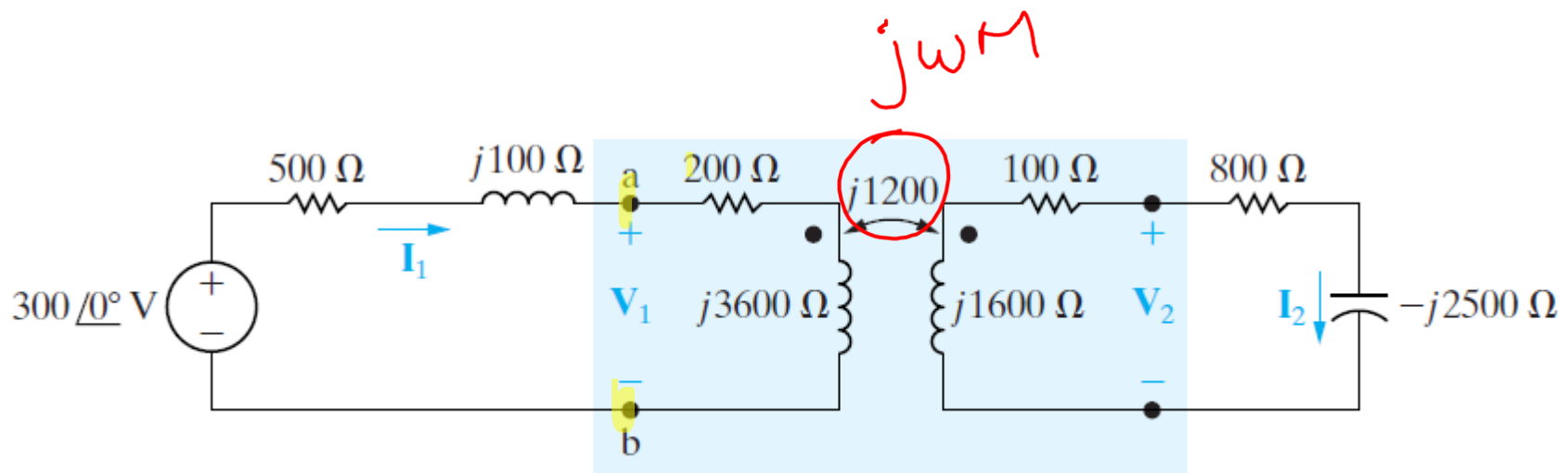
The parameters of a certain linear transformer are $R_1 = 200 \, \Omega$, $R_2 = 100 \, \Omega$, $L_1 = 9 \, \text{H}$, $L_2 = 4 \, \text{H}$, and $k = 0.5$. The transformer couples an impedance consisting of an $800 \, \Omega$ resistor in series with a $1 \, \mu\text{F}$ capacitor to a sinusoidal voltage source. The $300 \, \text{V}$ (rms) source has an internal impedance of $500 + j100 \, \Omega$ and a frequency of $400 \, \text{rad/s}$.



- Construct a frequency-domain equivalent circuit of the system.
- Calculate the self-impedance of the primary circuit.
- Calculate the self-impedance of the secondary circuit.
- Calculate the impedance reflected into the primary winding.
- Calculate the scaling factor for the reflected impedance.
- Calculate the impedance seen looking into the primary terminals of the transformer.

a, b

$$\begin{aligned}
 X_{L_1} &= j\omega L_1 = j(400)(9) = j3600 \, \Omega, \\
 X_{L_2} &= j\omega L_2 = j(400)(4) = j1600 \, \Omega, \\
 M &= 0.5\sqrt{(9)(4)} = 3 \, \text{H}, \\
 j\omega M &= j(400)(3) = j1200 \, \Omega, \\
 Z_L &\left\{ \begin{aligned} \frac{1}{j\omega C} &= \frac{10^6}{j400} = -j2500 \, \Omega. \end{aligned} \right.
 \end{aligned}$$



$$Z_{11} = 500 + j100 + 200 + j3600 = 700 + j3700 \, \Omega.$$

$$Z_{22} = 100 + j1600 + 800 - j2500 = 900 - j900 \, \Omega.$$

$$Z_r = \left(\frac{1200}{|900 - j900|} \right)^2 (900 + j900)$$

$$= \frac{8}{9} (900 + j900) = \underline{800 + j800 \, \Omega}.$$

$$Z_r = \frac{\omega^2 M^2}{|Z_{22}|^2} \times Z_{22}^*$$

The scaling factor by which Z_{22}^* is reflected is 8/9.

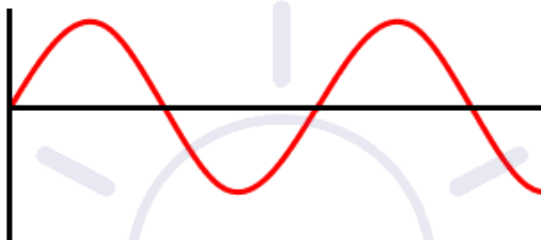
$$Z_{ab} = 200 + j3600 + \underline{800 + j800} = 1000 + j4400 \, \Omega.$$

Balanced Three Phase Circuits

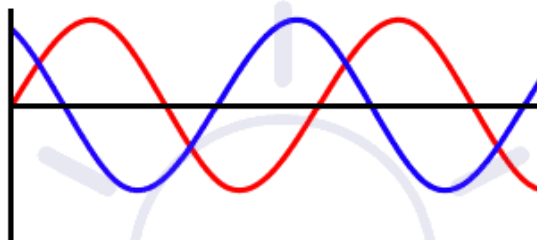


Why 3-Phase Power? Why Not 6- Φ , 12- Φ or More for Power Transmission?

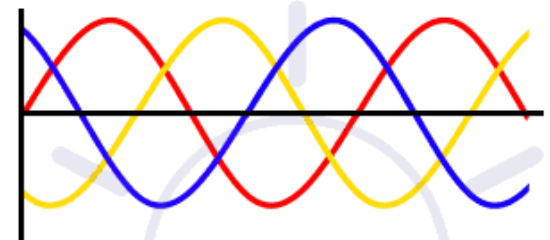
WWW.ELECTRICALTECHNOLOGY.ORG



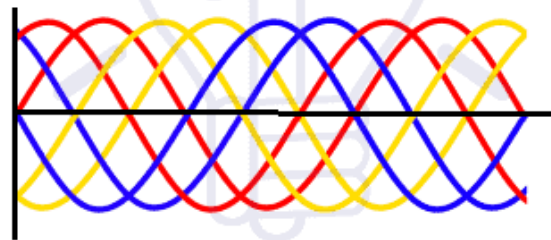
1 - Φ Supply



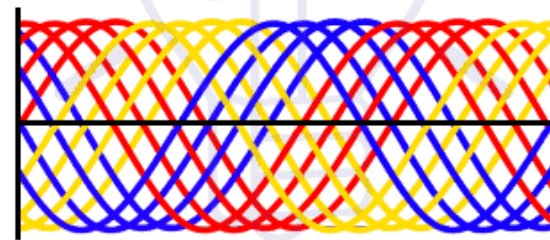
2 - Φ Supply



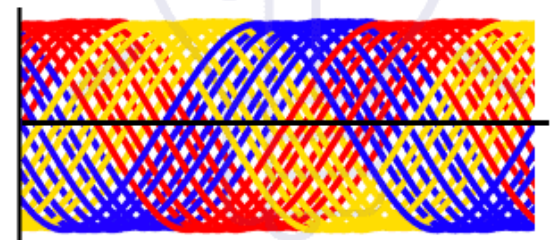
3 - Φ Supply



6 - Φ Supply



12 - Φ Supply



24 - Φ Supply

The main advantage of three phase over a single phase or two-phase system is that we can transmit constant and 1,732 (value of $\sqrt{3}$) times higher power.

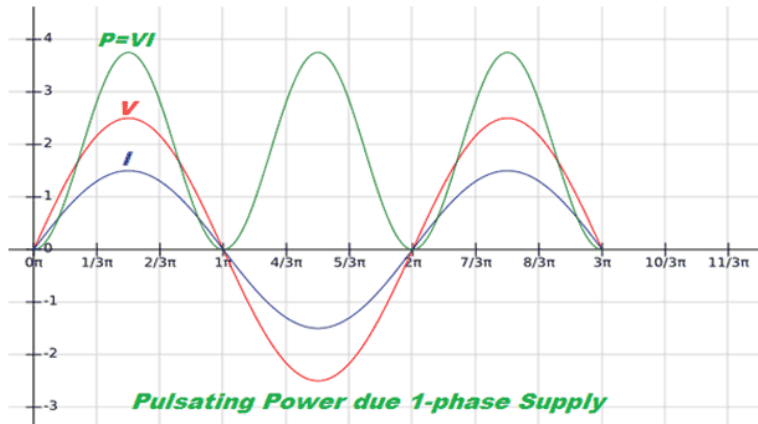


Figure (C)

Total power in single phase system is pulsating and so creates more vibrations.

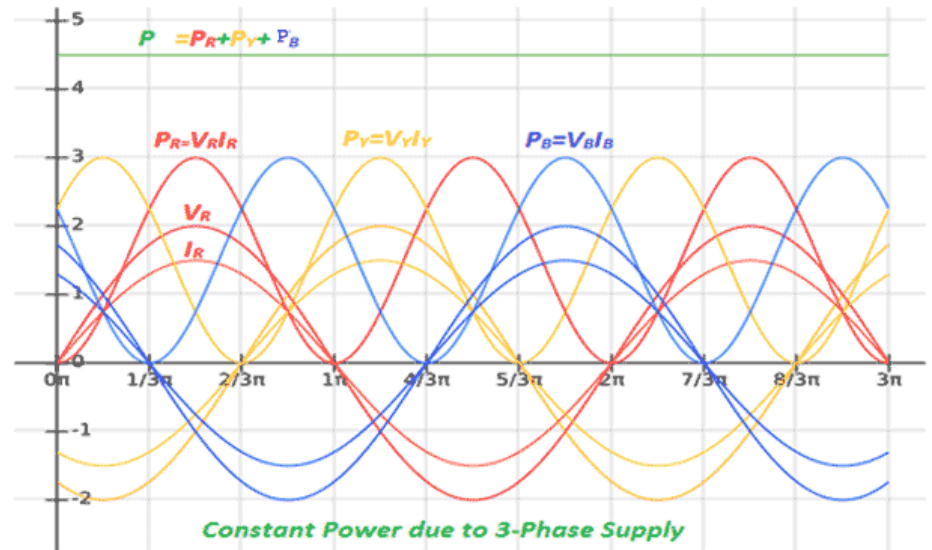
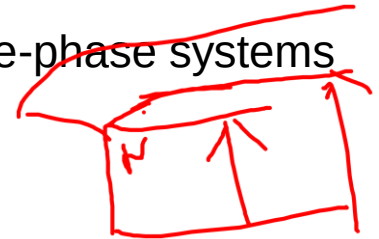


Figure (D)

Total power in three phase system is almost constant and so creates less vibration.

Why three phase power systems?

- For transmission and high-power applications:
 - Constant instantaneous power flow
 - Avoids the double frequency power oscillations of single-phase systems
 - Avoids torque oscillations of electrical machines
 - Effective utilization of conductors
 - A single-phase system requires two conductors
 - A three phase three wire system provides maximum utilization of the phases (i.e. three phases by three conductors) *2 Voltage values*
 - Two voltage levels available for three phase loads by changing between Wye and Delta connections
- Modern low voltage distribution systems in Norway
 - New installations: Three phase four wire systems
 - 230 V phase to neutral voltage utilized for single phase loads
 - 400 V line-to-line voltage for three phase loads
 - Usually Wye-connected loads that can be used with Delta-connection in 230 V three phase three-wire systems
 - Alternatively Delta-connected loads that can be used with Wye-connection in 690 V industrial power systems



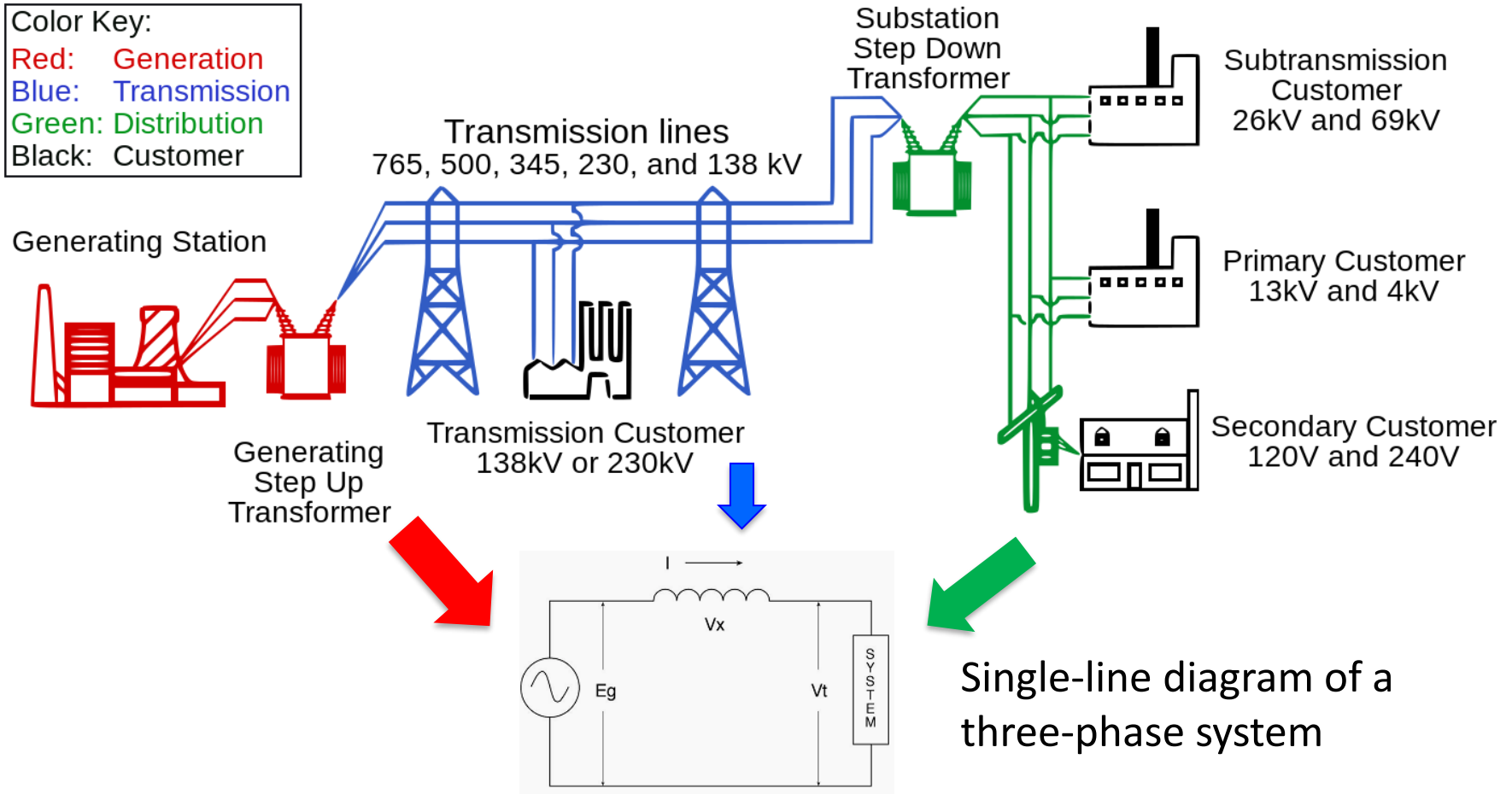
Objectives

- Know how to analyze a balanced, three-phase wye-wye (star-star) connected circuit
- Know how to analyze a balanced, three-phase wye-delta (star-delta) connected circuit
- Be able to calculate power (average, reactive, and complex) in any three-phase circuits.

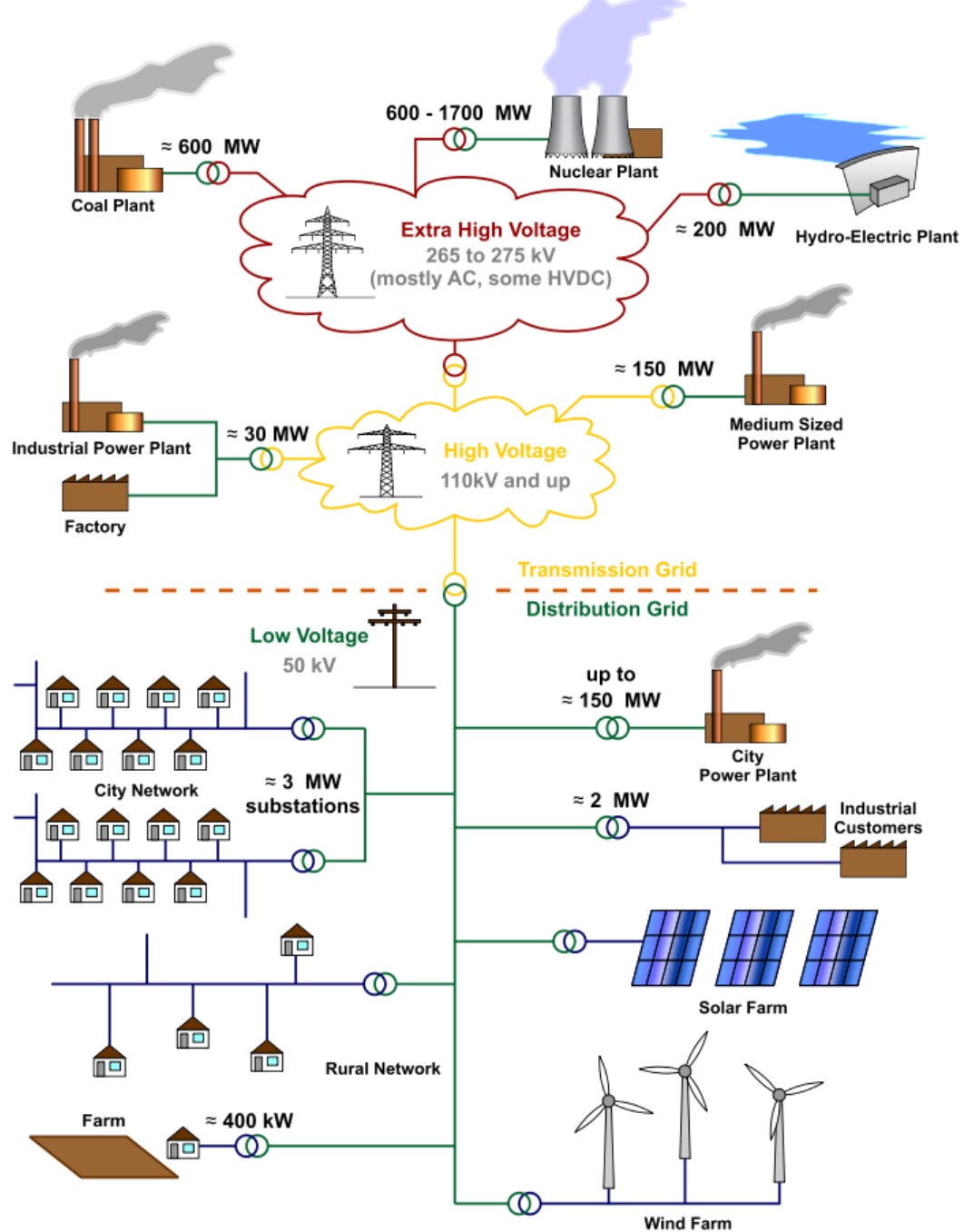


Three-Phase: Generating, transmitting, distributing and using large blocks of power

The Electric Power System



From generation to load: the equivalent circuit can be as above

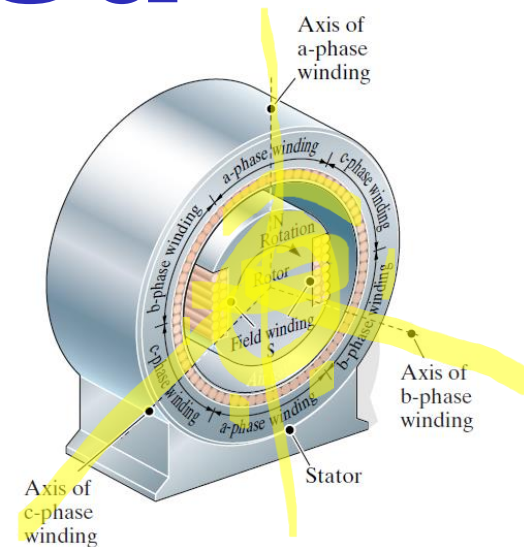


Centralized large scale generators: Nuclear, Coal, Hydro

Medium sized power plant at high voltage

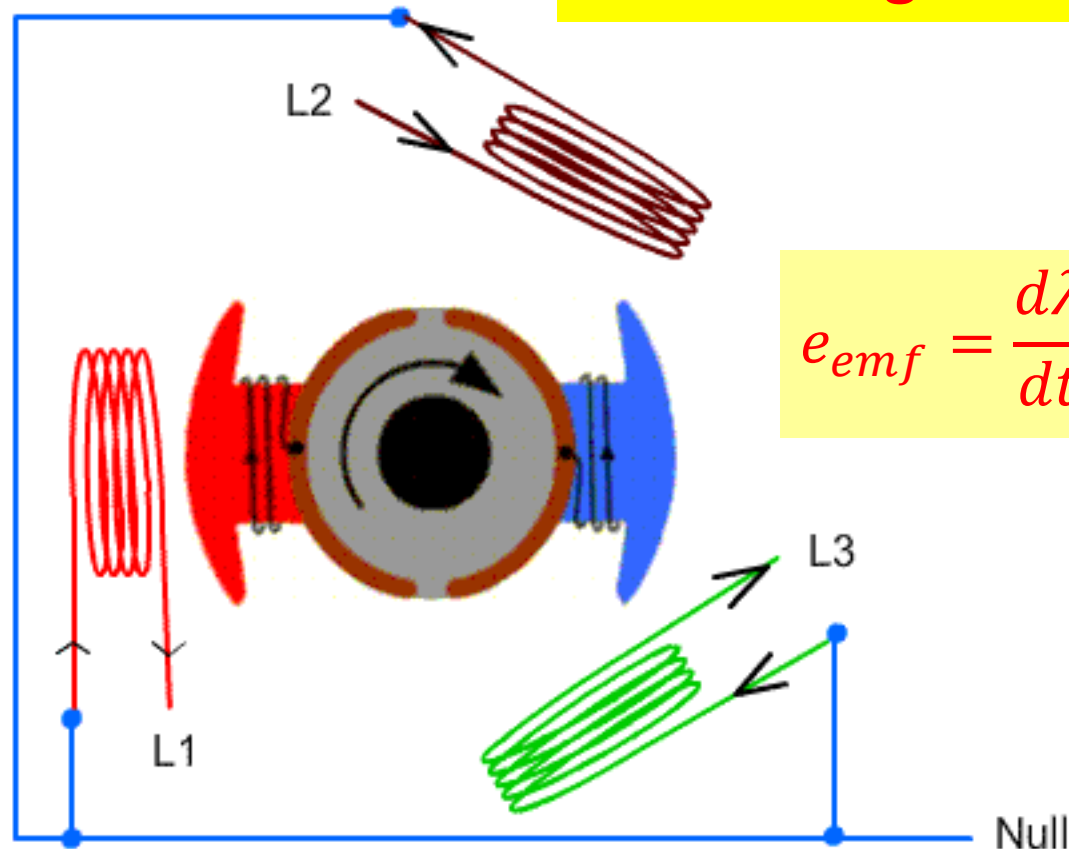
With dispersed and small scale generation at the distribution level: PV, Wind, etc

How the Three-phase Voltages are Generated



Three-Phase Voltage Source

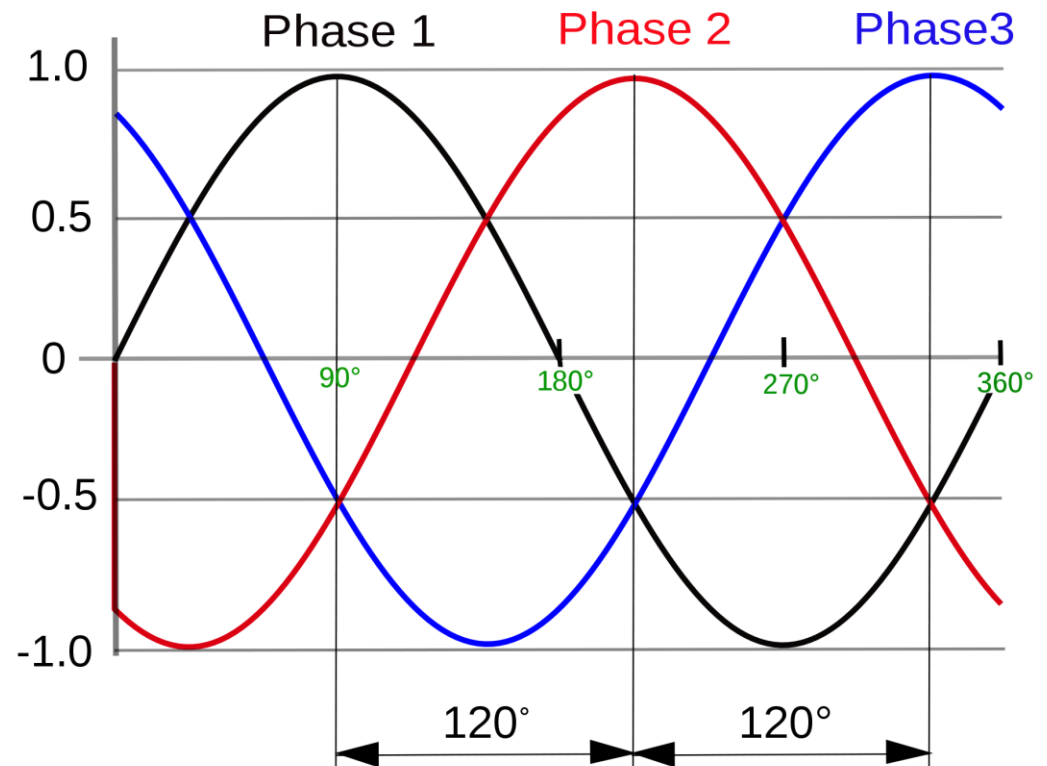
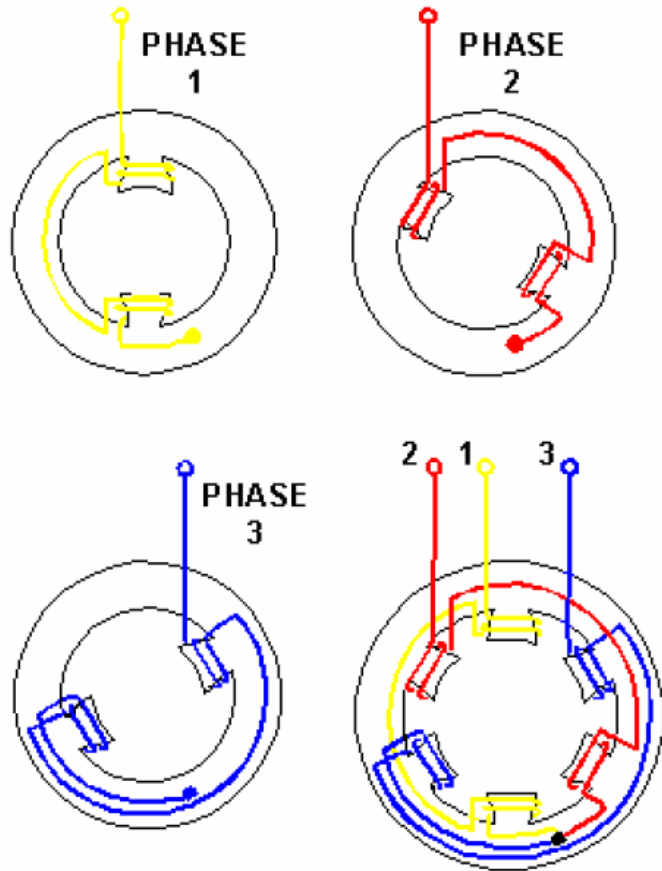
Electromagnetic Induction



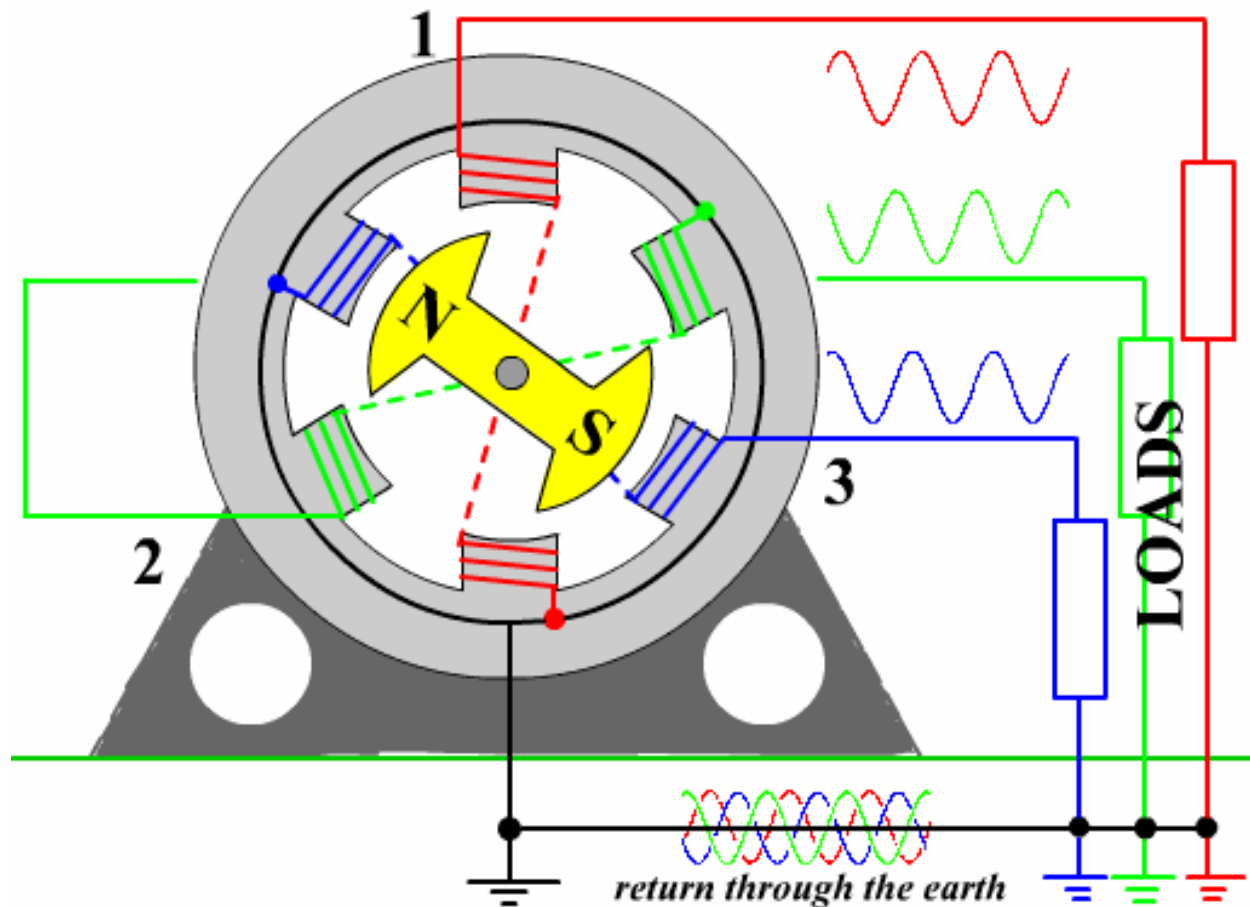
$$e_{emf} = \frac{d\lambda}{dt} = N \frac{d\psi}{dt}$$

<https://www.youtube.com/watch?v=tFaDfOap2fs>

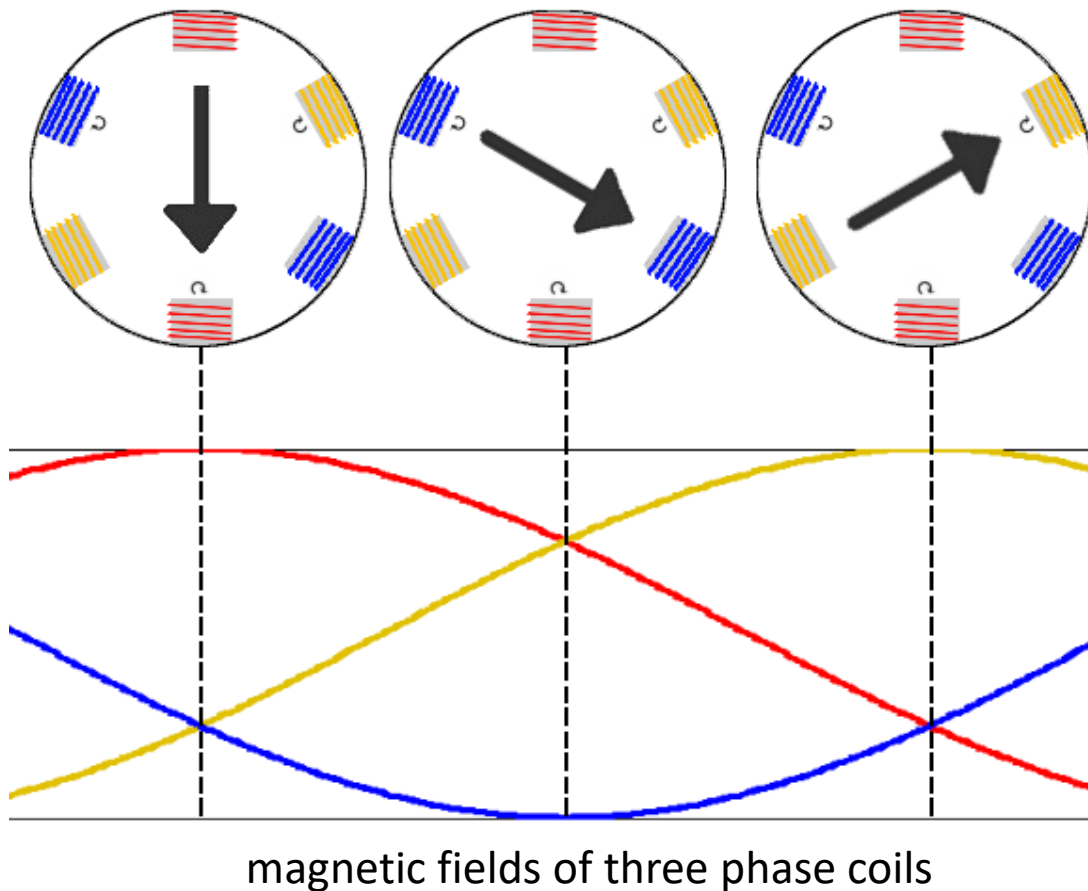
Three-phase balanced voltages



Three-phase sinusoidal voltage generator

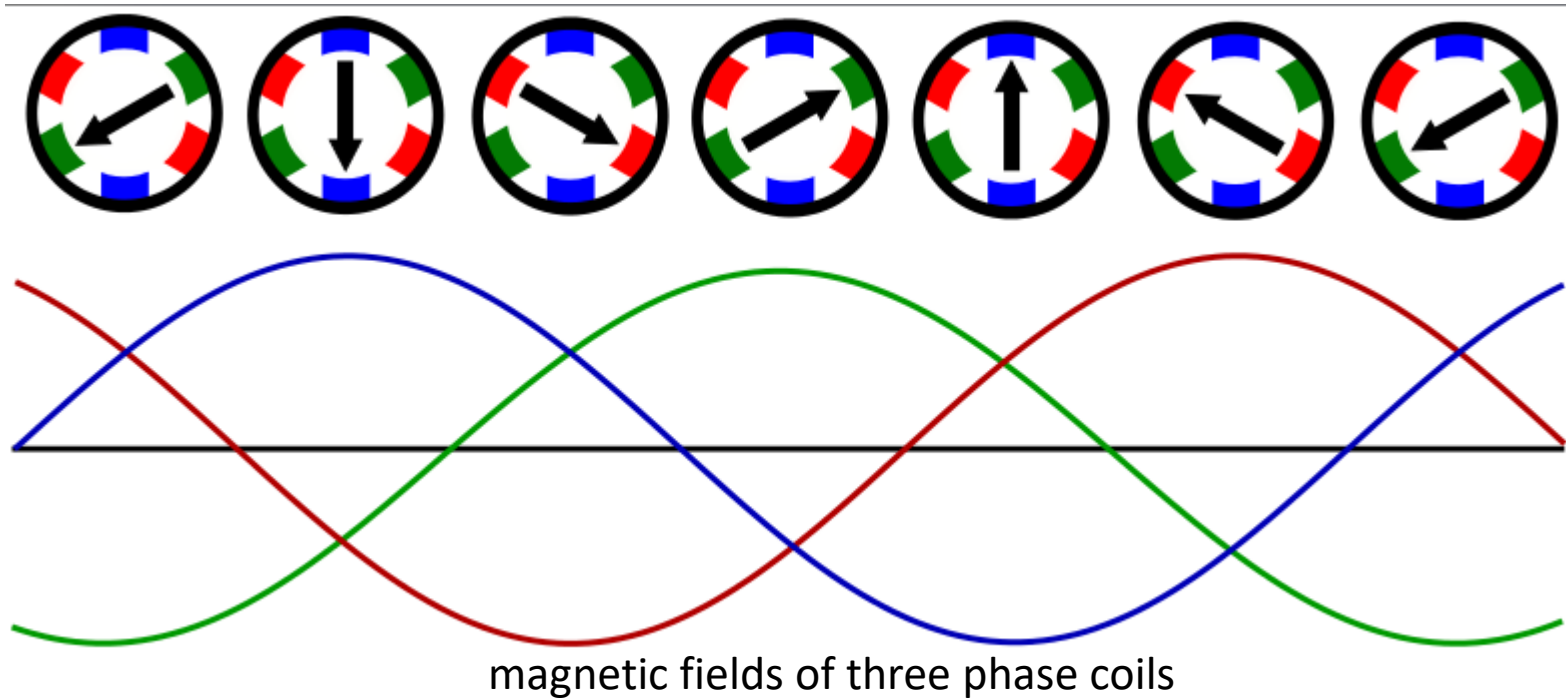


Generation of 3-phase sinusoidal voltage

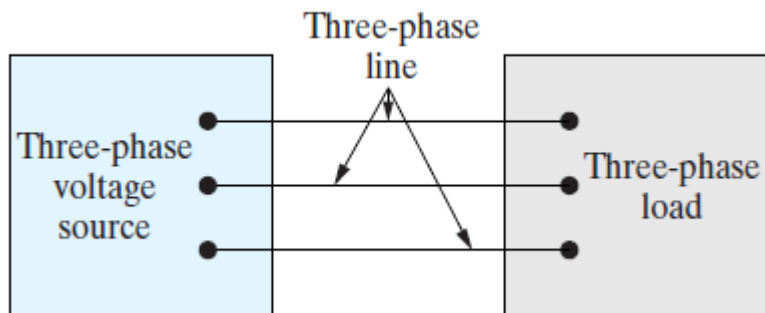


The black arrow denotes the rotating magnetic field created by the DC field magnetization of the rotor of a synchronous generator

Rotating magnetic field



Three-Phase Phasors



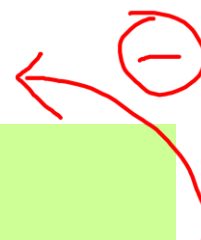
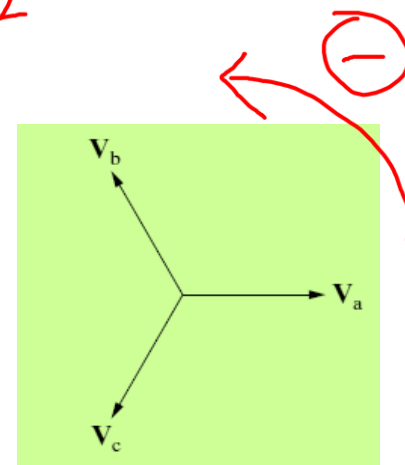
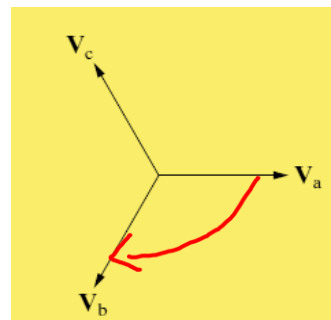
$$\begin{aligned} \mathbf{V}_a &= V_m \angle 0^\circ, \\ \mathbf{V}_b &= V_m \angle -120^\circ, \\ \mathbf{V}_c &= V_m \angle +120^\circ, \end{aligned}$$

$$\mathbf{V}_a + \mathbf{V}_b + \mathbf{V}_c = 0.$$

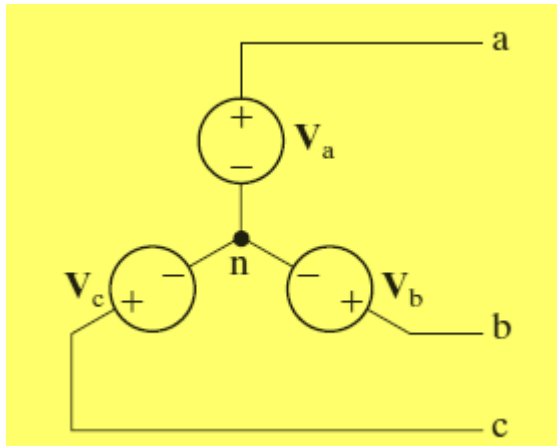
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$$v_a + v_b + v_c = 0.$$

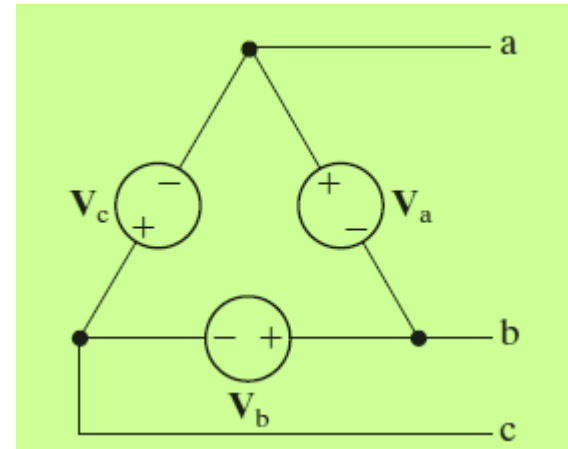
Positive Phase Sequence
Negative Phase Sequence



Y (star) and Δ (delta) connections



Also referred to as Wye

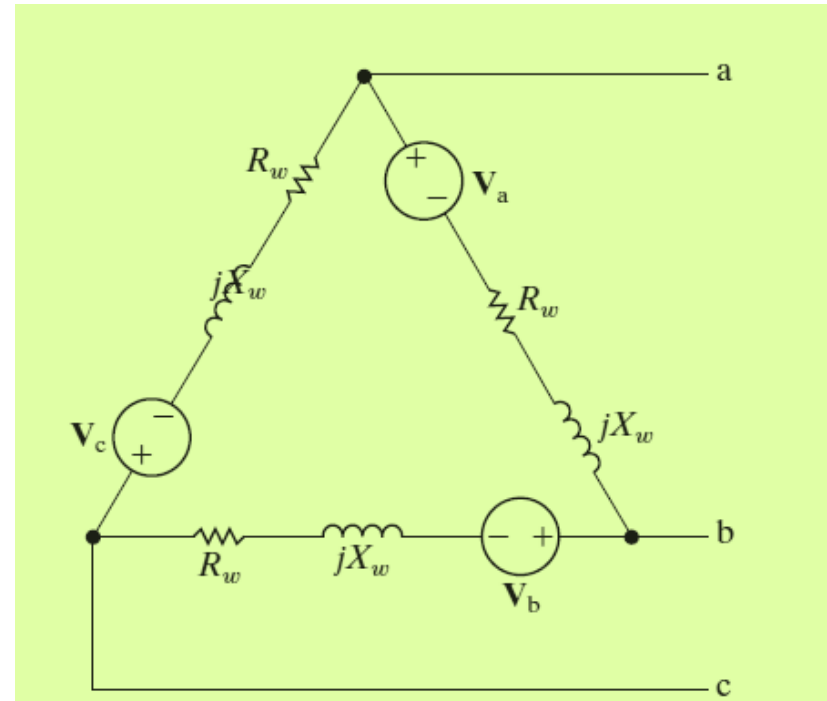
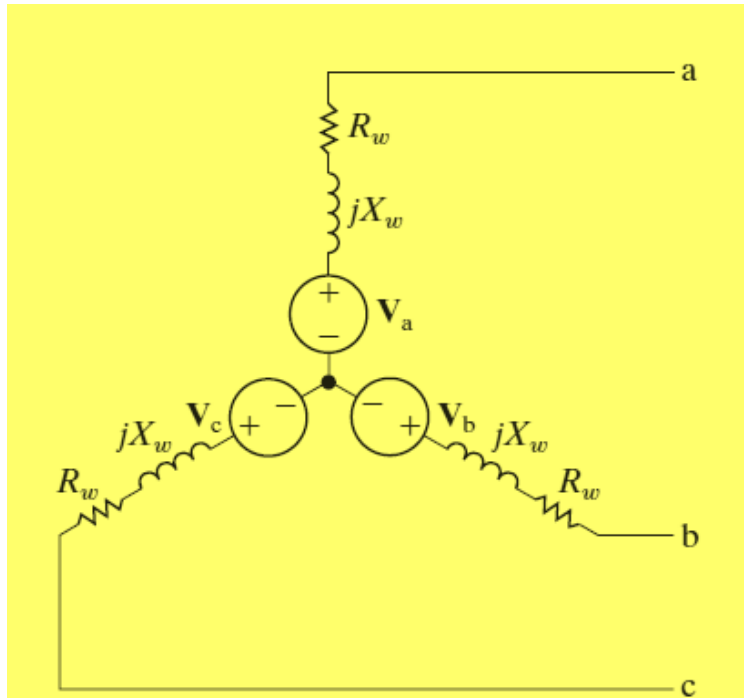


Source	Load
Y	Y
Y	Δ
Δ	Y
Δ	Δ

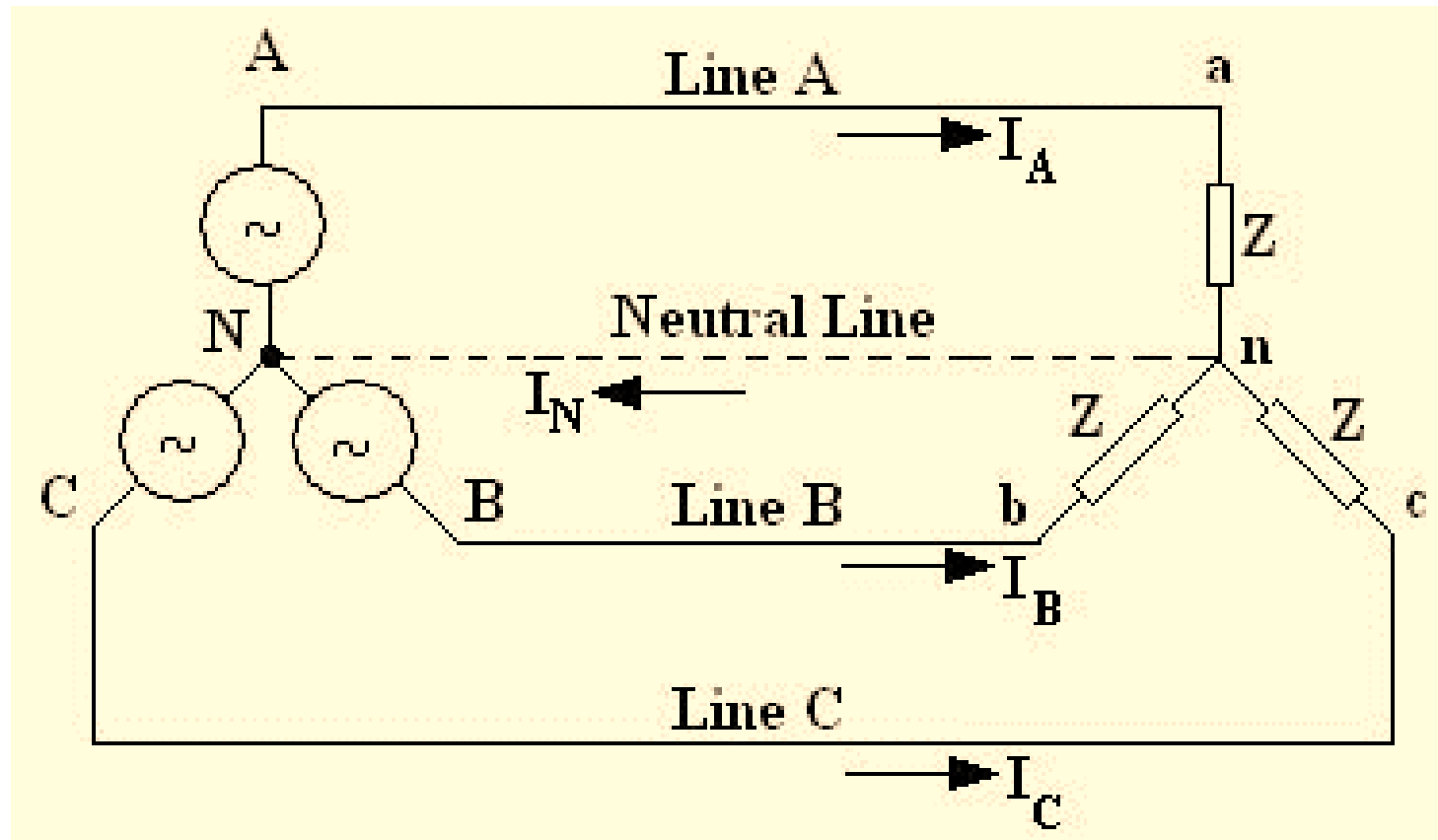
Y (star) and Δ (delta) connections

$$v_a + v_b + v_c = 0.$$

Y and Δ connections



Source – Load connection



Summary

- Introduced the **ideal transformer**: the **derived voltage and current ratio** are under the assumptions that there are no losses, no saturation and that magnetizing reluctance is negligible.
- Introduced the **linear transformer**: most of the assumptions of the ideal one do not hold, and the **main assumption is linear (magnetic behavior is linear $B=f(H)$)**.
- Developed **phasor circuits of magnetic coupling systems** and performed the **sinusoidal steady-state analysis**.
- **Fundamentals of balanced** three-phase system, knows how the three-phase voltages are generated.

End

https://www.youtube.com/watch?v=c9gm_NL7KyE